

PIZZA

SALES

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HELLO!

My Name is Jaysai Tandel

This project demonstrates SQL skills and techniques typically used by data analysts to explore, clean, and analyze retail sales data. The project involves setting up a pizza sales database, performing exploratory data analysis (EDA), and answering business questions through SQL queries.



DATA CLEANING

Fine Null Values

```
SELECT * FROM pizza_sales
WHERE
    pizza_id IS NULL OR order_id IS NULL
    OR pizza_name_id IS NULL OR quantity IS NULL
    OR order_date IS NULL OR order_time IS NULL
    OR unit_price IS NULL OR total_price IS NULL
    OR pizza_size IS NULL OR pizza_ingredients IS NULL
    OR pizza_name IS NULL;
```

DATA EXPLORATION



How many sales we have ?

```
SELECT COUNT(*) AS total_sales FROM pizza_sales
```

	total_sales
1	48620



DATA EXPLORATION



How many unique orders we have ?

```
SELECT COUNT(DISTINCT order_id) AS unique_orders  
FROM pizza_sales
```

	unique_orders
1	21350



DATA EXPLORATION



How many unique product categories in the dataset ?

```
SELECT DISTINCT pizza_category from pizza_sales
```

	pizza_category
1	Classic
2	Chicken
3	Veggie
4	Supreme



DATA ANALYSIS

retrieve all columns for sales made on '2015-10-03'

```
SELECT *  
FROM pizza_sales  
WHERE order_date = '2015-10-03'
```

	pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	unit_price	total_price	pizza_size	pizza_category	pizza_ingredients	pizza_name
1	37091	16378	calabrese_l	1	2015-10-03	11:53:41.0000000	20.25	20.25	L	Supreme	?duja Salami, Pancetta, Tomatoes, Red Onions, Friggi...	The Calabrese Pizza
2	37092	16379	pepperoni_m	1	2015-10-03	12:00:47.0000000	12.5	12.5	M	Classic	Mozzarella Cheese, Pepperoni	The Pepperoni Pizza
3	37093	16380	classic_dlx_s	1	2015-10-03	12:02:51.0000000	12	12	S	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, B...	The Classic Deluxe Pizza
4	37094	16380	ital_veggie_m	1	2015-10-03	12:02:51.0000000	16.75	16.75	M	Veggie	Eggplant, Artichokes, Tomatoes, Zucchini, Red Pepper...	The Italian Vegetables Pizza
5	37095	16381	napolitana_l	1	2015-10-03	12:04:45.0000000	20.5	20.5	L	Classic	Tomatoes, Anchovies, Green Olives, Red Onions, Garlic	The Napolitana Pizza
6	37096	16381	pepperoni_l	1	2015-10-03	12:04:45.0000000	15.25	15.25	L	Classic	Mozzarella Cheese, Pepperoni	The Pepperoni Pizza
7	37097	16382	big_meat_s	2	2015-10-03	12:09:45.0000000	12	24	S	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sausage	The Big Meat Pizza
8	37098	16382	five_cheese_l	1	2015-10-03	12:09:45.0000000	18.5	18.5	L	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked Goud...	The Five Cheese Pizza
9	37099	16383	pepperoni_s	1	2015-10-03	12:22:07.0000000	9.75	9.75	S	Classic	Mozzarella Cheese, Pepperoni	The Pepperoni Pizza
10	37100	16384	classic_dlx_m	1	2015-10-03	12:29:09.0000000	16	16	M	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, B...	The Classic Deluxe Pizza

DATA ANALYSIS



retrieve all transactions where the Pizza category is 'Classic' in the month of Nov-2015

```
SELECT * FROM pizza_sales
WHERE pizza_category = 'Classic'
AND FORMAT(order_date, 'yyyy-MM') = '2015-11'
```

	pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	unit_price	total_price	pizza_size	pizza_category	pizza_ingredients	pizza_name
1	40578	17880	big_meat_s	1	2015-11-01	12:12:20.0000000	12	12	S	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sausage	The Big Meat Pizza
2	40582	17880	hawaiian_m	1	2015-11-01	12:12:20.0000000	13.25	13.25	M	Classic	Sliced Ham, Pineapple, Mozzarella Cheese	The Hawaiian Pizza
3	40583	17880	hawaiian_s	1	2015-11-01	12:12:20.0000000	10.5	10.5	S	Classic	Sliced Ham, Pineapple, Mozzarella Cheese	The Hawaiian Pizza
4	40584	17880	napolitana_m	1	2015-11-01	12:12:20.0000000	16	16	M	Classic	Tomatoes, Anchovies, Green Olives, Red Onions, Garlic	The Napolitana Pizza
5	40585	17880	pepperoni_m	1	2015-11-01	12:12:20.0000000	12.5	12.5	M	Classic	Mozzarella Cheese, Pepperoni	The Pepperoni Pizza
6	40587	17881	ital_cpello_s	1	2015-11-01	12:23:50.0000000	12	12	S	Classic	Capocollo, Red Peppers, Tomatoes, Goat Cheese, Gar...	The Italian Capocollo Pizza
7	40588	17882	classic_dlx_m	1	2015-11-01	12:31:08.0000000	16	16	M	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, B...	The Classic Deluxe Pizza
8	40589	17883	big_meat_s	1	2015-11-01	12:38:50.0000000	12	12	S	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sausage	The Big Meat Pizza
9	40593	17886	classic_dlx_l	1	2015-11-01	13:02:02.0000000	20.5	20.5	L	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, B...	The Classic Deluxe Pizza
10	40601	17887	napolitana_m	1	2015-11-01	13:09:02.0000000	16	16	M	Classic	Tomatoes, Anchovies, Green Olives, Red Onions, Garlic	The Napolitana Pizza

DATA ANALYSIS

calculate the total sales (total_sale) for each Pizza category.

```
SELECT pizza_category,  
       SUM(CAST(total_price AS DECIMAL(10,2))) AS net_sale,  
       COUNT(*) AS total_orders  
FROM pizza_sales  
GROUP BY pizza_category;
```

	pizza_category	net_sale	total_orders
1	Classic	220053.10	14579
2	Chicken	195919.50	10815
3	Veggie	193690.45	11449
4	Supreme	208197.00	11777

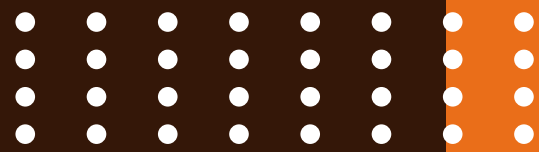
DATA ANALYSIS

Find the average unit price of customers who purchased items from the 'Veggie' category.

```
SELECT  
    ROUND(AVG(unit_price),2) AS avg_unit_price  
FROM pizza_sales  
WHERE pizza_category = 'Veggie'
```

	avg_unit_price
1	16.61

DATA ANALYSIS



Find all transactions where the total_price is greater than 15.

```
SELECT * FROM pizza_sales
WHERE total_price > 15
```

SQL Message

	pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	unit_price	total_price	pizza_size	pizza_category	pizza_ingredients	pizza_name
1	2	2	classic_dlx_m	1	2015-01-01	11:57:40.0000000	16	16	M	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, B...	The Classic Deluxe Pizza
2	3	2	five_cheese_l	1	2015-01-01	11:57:40.0000000	18.5	18.5	L	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked Goud...	The Five Cheese Pizza
3	4	2	ital_supr_l	1	2015-01-01	11:57:40.0000000	20.75	20.75	L	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza
4	5	2	mexicana_m	1	2015-01-01	11:57:40.0000000	16	16	M	Veggie	Tomatoes, Red Peppers, Jalapeno Peppers, Red Onio...	The Mexicana Pizza
5	6	2	thai_ckn_l	1	2015-01-01	11:57:40.0000000	20.75	20.75	L	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, Thai Sw...	The Thai Chicken Pizza
6	7	3	ital_supr_m	1	2015-01-01	12:12:28.0000000	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza
7	8	3	prsc_argla_l	1	2015-01-01	12:12:28.0000000	20.75	20.75	L	Supreme	Prosciutto di San Daniele, Arugula, Mozzarella Cheese	The Prosciutto and Arugula Pizza
8	9	4	ital_supr_m	1	2015-01-01	12:16:31.0000000	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza
9	10	5	ital_supr_m	1	2015-01-01	12:21:30.0000000	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza
10	17	9	ital_cpcllo_l	1	2015-01-01	12:52:01.0000000	20.5	20.5	L	Classic	Capocollo, Red Peppers, Tomatoes, Goat Cheese, Gar...	The Italian Capocollo Pizza

DATA ANALYSIS

Find the total number of transactions (order_id) made by each category.

```
SELECT pizza_category,  
       COUNT(*) AS total_transaction  
FROM pizza_sales  
GROUP BY pizza_category;
```

	pizza_category	total_transaction
1	Classic	14579
2	Chicken	10815
3	Veggie	11449
4	Supreme	11777

DATA ANALYSIS

Calculate the average sale for each month. Find out best selling month in each year

```
WITH monthly_sales AS (  
  SELECT  
    YEAR(order_date) AS year,  
    MONTH(order_date) AS month,  
    AVG(total_price) AS avg_sale  
  FROM pizza_sales  
  GROUP BY YEAR(order_date), MONTH(order_date)  
)  
SELECT year, month, avg_sale  
FROM (  
  SELECT  
    year,  
    month,  
    avg_sale,  
    RANK() OVER (PARTITION BY year ORDER BY avg_sale DESC) AS rank  
  FROM monthly_sales  
) t1  
WHERE rank = 1;
```

	year	month	avg_sale
1	2015	6	16.9516025007734

DATA ANALYSIS

Find the top 5 pizza name based on the highest total sales

```
SELECT TOP 5
    pizza_name,
    SUM(total_price) as total_sales
FROM pizza_sales
GROUP BY pizza_name
ORDER BY SUM(total_price) DESC
```

	pizza_name	total_sales
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768
3	The California Chicken Pizza	41409.5
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25

DATA ANALYSIS

Find the number of unique customers who purchased items from each category.

```
SELECT
    pizza_category,
    COUNT(DISTINCT order_id) AS cnt_unique_cs
FROM pizza_sales
GROUP BY pizza_category;
```

	pizza_category	cnt_unique_cs
1	Classic	10859
2	Chicken	8536
3	Veggie	8941
4	Supreme	9085

DATA ANALYSIS

Create each shift and number of orders (Example Morning <12, Afternoon Between 12 & 17, Evening >17)

```
WITH hourly_sale AS (  
  SELECT *,  
    CASE  
      WHEN DATEPART(HOUR, order_time) < 12 THEN 'Morning'  
      WHEN DATEPART(HOUR, order_time) BETWEEN 12 AND 17 THEN 'Afternoon'  
      ELSE 'Evening'  
    END AS shift  
  FROM pizza_sales  
)  
SELECT  
  shift,  
  SUM(total_price) AS total_sales  
FROM hourly_sale  
GROUP BY shift;
```

	shift	total_sales
1	Evening	286107.450283051
2	Morning	45322.4500484467
3	Afternoon	486430.150506973

**THANK YOU
FOR ATTENTION**

